

Practice set- 1: convex set

2.1 Let $C \subseteq \mathbb{R}^n$ be a convex set, with $x_1, \dots, x_k \in C$, and let $\theta_1, \dots, \theta_k \in \mathbb{R}$ satisfy $\theta_i \geq 0$, $\theta_1 + \dots + \theta_k = 1$. Show that $\theta_1 x_1 + \dots + \theta_k x_k \in C$. (The definition of convexity is that this holds for $k = 2$; you must show it for arbitrary k .) Hint. Use induction on k .

2.2 Show that a set is convex if and only if its intersection with any line is convex. Show that a set is affine if and only if its intersection with any line is affine.

2.8 Which of the following sets S are polyhedra?

If possible, express S in the form $S = \{x \mid Ax \leq b, Fx = g\}$.

(a) $S = \{y_1 a_1 + y_2 a_2 \mid -1 \leq y_1 \leq 1, -1 \leq y_2 \leq 1\}$, where $a_1, a_2 \in \mathbb{R}^n$.

2.12 Which of the following sets are convex?

(a) A slab, i.e., a set of the form $\{x \in \mathbb{R}^n \mid \alpha \leq a^T x \leq \beta\}$.

(b) A rectangle, i.e., a set of the form $\{x \in \mathbb{R}^n \mid \alpha_i \leq x_i \leq \beta_i, i = 1, \dots, n\}$. A rectangle is sometimes called a hyperrectangle when $n > 2$.

(c) A wedge, i.e., $\{x \in \mathbb{R}^n \mid a_1^T x \leq b_1, a_2^T x \leq b_2\}$

2.11 Hyperbolic sets. Show that the hyperbolic set $\{x \in \mathbb{R}^2_+ \mid x_1 x_2 \geq 1\}$ is convex. As a generalization, show that $\{x \in \mathbb{R}^n_+ \mid \prod_{i=1}^n x_i \geq 1\}$ is convex.

(Hint. If $a, b \geq 0$ and $0 \leq \theta \leq 1$, then $a^\theta b^{1-\theta} \leq \theta a + (1 - \theta)b$; see §3.1.9. 5)

2.15 Some sets of probability distributions. Let x be a real-valued random variable with $\text{prob}(x = a_i) = p_i$, $i = 1, \dots, n$, where $a_1 < a_2 < \dots < a_n$. Ofcourse $p \in \mathbb{R}^n$ lies

in the standard probability simplex $P = \{p \mid 1^T p = 1, p \geq 0\}$. Which of the following conditions are convex in p ? (That is, for which of the following conditions is the set of $p \in P$ that satisfy the condition convex?)

(a) $\alpha \leq E f(x) \leq \beta$, where $E f(x)$ is the expected value of $f(x)$, i.e., $E f(x) = \sum_{i=1}^n p_i f(a_i)$. (The function $f: \mathbb{R} \rightarrow \mathbb{R}$ is given.)

(b) $\text{prob}(x > \alpha) \leq \beta$.

(c) $E |x_3| \leq \alpha E |x|$.

(d) $E x^2 \leq \alpha$.

(e) $E x^2 \geq \alpha$.

(f) $\text{var}(x) \leq \alpha$, where $\text{var}(x) = E (x - E x)^2$ is the variance of x .

(g) $\text{var}(x) \geq \alpha$.

(h) $\text{quartile}(x) \geq \alpha$, where $\text{quartile}(x) = \inf \{\beta \mid \text{prob}(x \leq \beta) \geq 0.25\}$.

(i) $\text{quartile}(x) \leq \alpha$

2.21 The set of separating hyperplanes. Suppose that C and D are disjoint subsets of $\mathbb{R}^n$. Consider the set of $(a, b) \in \mathbb{R}^{n+1}$ for which $a^T x \leq b$ for all $x \in C$, and $a^T x \geq b$ for all $x \in D$. Show that this set is a convex cone (which is the singleton $\{0\}$ if there is no hyperplane that separates C and D).

2.24 Supporting hyperplanes.

(a) Express the closed convex set $\{x \in \mathbb{R}^2_+ \mid x_1 x_2 \geq 1\}$ as an intersection of half-spaces.